

Multi-Period Traffic Folding for Efficient IoT Intrusion Detection

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Abstract. Internet of Things (IoT) intrusion detection requires accurate attack recognition with limited computation. Existing deep IDS models usually treat traffic as a one-dimensional sequence, which may weaken periodic communication patterns, repeated attack bursts, and stage-wise behavior changes. This paper proposes MPF-LiteIDS, a compact period-aware network for IoT intrusion detection. MPF-LiteIDS first discovers dominant frequency indices from flow-level traffic and folds one-dimensional sequences into multiple two-dimensional periodic maps. This representation models intra-period variations and inter-period shifts jointly. A depthwise separable convolution module extracts compact periodic features, and period-stage attention emphasizes discriminative periods and traffic stages. Self-distillation is used during training to improve the compact detector without increasing inference cost. Experiments on three IoT intrusion detection datasets show that MPF-LiteIDS outperforms strong baselines, including Transformer and TimesNet. On BoT-IoT, MPF-LiteIDS achieves a 97.81% F1-score and reduces parameters by 36.4% compared with CNN-BiLSTM-Attention.

Keywords: IoT security · intrusion detection · multi-period traffic folding · lightweight neural network · attention mechanism · self-distillation

1 Introduction

Internet of Things (IoT) networks generate heterogeneous traffic from resource-constrained devices, lightweight protocols, and dynamic services. Attacks such as DoS, DDoS, reconnaissance, and data theft often appear as bursty, repetitive, or stage-wise traffic patterns. Therefore, an IoT intrusion detection system (IDS) should achieve accurate detection while remaining efficient for edge-oriented deployment [1, 4].

Existing IDS methods mainly include traditional machine learning and deep learning models. Tree-based ensemble methods are efficient for flow-level features, but they depend on handcrafted representations and have limited temporal modeling ability [3, 5, 11]. Deep models, such as CNN, LSTM, BiLSTM, and attention-based networks, improve feature learning from traffic sequences [1, 15]. However, most of them still model IoT traffic as a flat one-dimensional sequence,

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which may weaken repeated attack bursts, periodic device communication, and stage-wise behavior shifts.

A key observation is that IoT traffic often contains multiple dominant periods. Period-aware time-series models show that reorganizing one-dimensional sequences into structured representations can better expose temporal variations [7, 16]. However, directly applying generic period-aware backbones to IoT IDS may introduce unnecessary computation and may ignore flow-level traffic characteristics.

To address these issues, we propose *MPF-LiteIDS*, a period-aware traffic folding network for lightweight IoT intrusion detection. It discovers dominant frequency indices, converts them into selected periods, and folds one-dimensional flow features into two-dimensional periodic maps. A depthwise separable convolution extractor learns compact local patterns [6, 10], period-stage attention emphasizes discriminative periods and stages, and self-distillation improves the compact detector without an external teacher [8, 9].

The main contributions are summarized as follows:

- We propose a multi-period traffic folding module that transforms one-dimensional flow sequences into two-dimensional periodic maps for modeling intra-period attack bursts and inter-period behavior shifts.
- We design a lightweight period-stage detection network with depthwise separable convolution, BiLSTM encoding, and period-stage attention to improve the accuracy-efficiency trade-off.
- We evaluate MPF-LiteIDS on three IoT intrusion detection datasets against strong baselines, including Transformer [15] and TimesNet [16]. Comparative experiments, ablation study, complexity analysis, and attention visualization verify its effectiveness.

2 Related Work

2.1 Learning-Based IoT Intrusion Detection

Learning-based methods have been widely used in IoT intrusion detection because they can identify attack patterns from high-dimensional traffic features [1, 4]. Traditional machine learning models are efficient for tabular flow features but depend on feature engineering and usually lack explicit temporal modeling ability. Deep models further use convolutional, recurrent, and attention-based structures to capture local correlations and temporal dependencies. However, many existing deep IDS models still process traffic as a one-dimensional sequence, making it difficult to explicitly represent repeated periodic behaviors and stage-wise attack evolution.

2.2 Period-Aware Time-Series Modeling

Period-aware representation learning provides an effective way to model complex temporal patterns [7, 16]. Instead of analyzing a sequence only in its original

form, such methods identify dominant periods and reorganize temporal signals into structured representations, enabling intra-period variations and inter-period changes to be modeled jointly. However, generic period-aware backbones are not directly optimized for IoT intrusion detection, where flow-level features, mixed normal/attack behaviors, and imbalanced categories are common. This motivates a traffic-oriented folding strategy for exposing periodic attack structures with limited complexity.

2.3 Lightweight Detection for Edge Deployment

Efficiency is critical for IoT IDS because edge devices usually have limited memory, computation, and latency budgets. Depthwise separable convolution reduces parameters and FLOPs by decomposing standard convolution into depthwise and pointwise operations [6, 10]. Knowledge distillation can improve compact models, but conventional distillation often requires a large teacher network [9]. Self-distillation introduces soft training supervision without changing the inference path [8]. These studies motivate our lightweight period-stage detector with compact convolution, period-aware fusion, and self-distillation.

3 Proposed Method

3.1 Overview of MPF-LiteIDS

MPF-LiteIDS targets two practical limitations of existing deep IDS models: one-dimensional sequence modeling may miss periodic attack bursts, while heavy hybrid networks often increase computation. Fig. 1 shows the overall architecture.

Given an input traffic sequence $X \in \mathbb{R}^{B \times T \times F}$, MPF-LiteIDS first discovers dominant periods and folds the sequence into multi-period two-dimensional traffic maps. The folded maps are processed by a lightweight periodic feature extractor, BiLSTM temporal encoder, and period-stage attention module. The fused representation is finally sent to a classifier, while self-distillation is used only during training.

3.2 Multi-Period Traffic Folding

IoT traffic often contains repeated communication intervals, periodic sensing behaviors, and bursty attack responses. Directly modeling such traffic as a flat sequence may hide the interaction between short-term intra-period changes and long-term inter-period shifts. To explicitly expose these structures, MPF-LiteIDS introduces a multi-period traffic folding module, as shown in Fig. 2.

The amplitude spectrum is computed along the temporal dimension of each traffic window:

$$A(q) = \frac{1}{BF} \sum_{b=1}^B \sum_{f=1}^F |\text{FFT}(X_{b,:,f})(q)|, \quad (1)$$

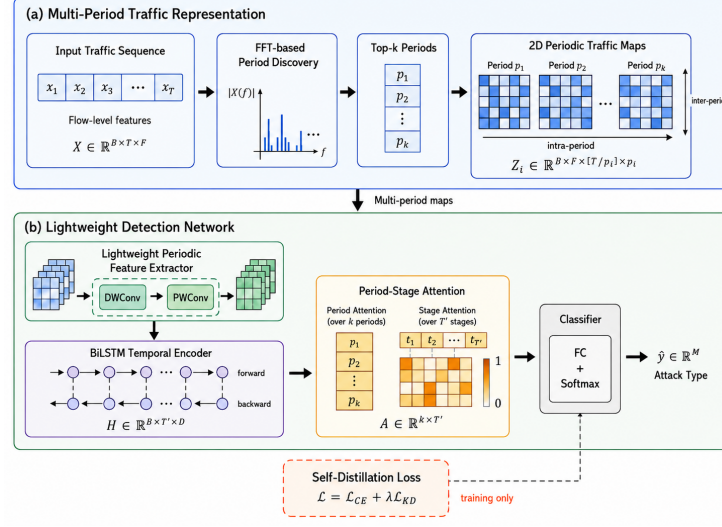


Fig. 1. Overall architecture of MPF-LiteIDS.

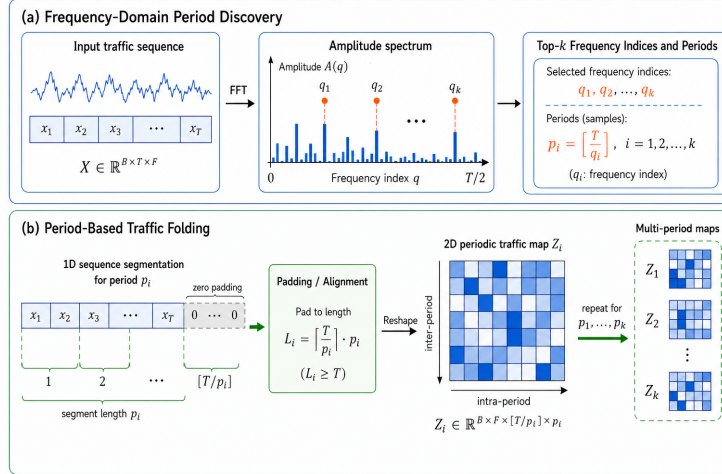


Fig. 2. Multi-period traffic folding process.

where $q = 1, \dots, \lfloor T/2 \rfloor$ denotes the non-zero frequency index. The top- k dominant frequency indices are selected as

$$\{q_1, \dots, q_k\} = \text{TopK}(A(q)). \quad (2)$$

Each selected frequency index q_i is converted into a traffic period:

$$p_i = \left\lceil \frac{T}{q_i} \right\rceil, \quad i = 1, \dots, k. \quad (3)$$

Since T may not be divisible by p_i , the sequence is padded to $L_i = \lceil T/p_i \rceil p_i$ and reshaped into

$$Z_i \in \mathbb{R}^{B \times F \times \lceil T/p_i \rceil \times p_i}. \quad (4)$$

In Z_i , the last dimension describes intra-period variation, while the third dimension describes inter-period variation. Applying the same operation to all selected periods produces

$$\mathcal{Z} = \{Z_1, Z_2, \dots, Z_k\}. \quad (5)$$

Compared with one-dimensional traffic modeling, this representation allows the detector to observe both local attack bursts within each period and behavioral shifts across repeated periods.

3.3 Lightweight Period-Stage Detection

For each periodic map Z_i , MPF-LiteIDS uses depthwise separable convolution to extract compact local patterns with lower computation:

$$S_i = \phi(\text{PWConv}(\text{DWConv}(Z_i))), \quad (6)$$

where $\text{DWConv}(\cdot)$ and $\text{PWConv}(\cdot)$ denote depthwise and pointwise convolution, and $\phi(\cdot)$ denotes normalization and nonlinear activation.

The extracted features are encoded by a BiLSTM:

$$H_i = \text{BiLSTM}(S_i), \quad H_i \in \mathbb{R}^{B \times T' \times D}, \quad (7)$$

where T' is the reduced temporal length and D is the hidden dimension. Features from all selected periods are stacked as $H \in \mathbb{R}^{B \times k \times T' \times D}$.

To select discriminative periods and stages, MPF-LiteIDS computes period-stage attention:

$$\alpha_{i,t} = \frac{\exp(v^\top \tanh(WH_{i,t} + b))}{\sum_{m=1}^k \sum_{n=1}^{T'} \exp(v^\top \tanh(WH_{m,n} + b))}. \quad (8)$$

The final representation is obtained by

$$\tilde{H} = \sum_{i=1}^k \sum_{t=1}^{T'} \alpha_{i,t} H_{i,t}, \quad (9)$$

and the prediction is generated as

$$\hat{y} = \text{Softmax}(W_c \tilde{H} + b_c). \quad (10)$$

The training objective combines classification loss and self-distillation loss:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{SD}, \quad (11)$$

where $\mathcal{L}_{CE}$ is the cross-entropy loss, $\mathcal{L}_{SD}$ aligns the softened prediction distribution with its stop-gradient counterpart, and λ controls the distillation strength. Since self-distillation is used only during training, the inference path remains unchanged.

4 Experiments

4.1 Datasets and Preprocessing

Experiments are conducted on three public intrusion detection datasets: BoT-IoT [13], ToN-IoT [2], and CICIoT2023 [14]. These datasets contain different IoT and network traffic scenarios, including DoS, DDoS, reconnaissance, scanning, brute-force attacks, and data theft. To ensure a consistent evaluation protocol, all datasets are preprocessed using the same procedure. Rare fine-grained labels are merged into high-level attack families when necessary. The sample numbers in Table 1 denote the processed subsets used in our experiments after cleaning, label merging, and stratified sampling.

Table 1. Summary of datasets used in the experiments.

Dataset	Proc. Samples	Features	Classes	Split
BoT-IoT	1,000,000	43	5	7:1:2
ToN-IoT	461,043	39	10	7:1:2
CICIoT2023	1,200,000	46	8	7:1:2

For each dataset, duplicated records, missing values, and invalid entries are removed. Categorical attributes are encoded into numerical values. Numerical features are normalized by min-max normalization:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \quad (12)$$

After preprocessing, flow records are sorted according to their original temporal order. To avoid overlap-induced data leakage, the data are first divided into training, validation, and testing subsets with a ratio of 70%, 10%, and 20%, respectively. Sliding-window segmentation is then performed independently within each subset. Each window contains T consecutive flow records with stride s , and

the label of a window is assigned according to the last record in the window. Unless otherwise specified, we set $T = 64$ and $s = 32$. In this way, windows from different subsets do not share flow records. The validation set is used for hyperparameter selection and early stopping, while all reported results are calculated on the testing set.

4.2 Compared Methods

MPF-LiteIDS is compared with three groups of baselines: classical machine learning models, including RF [3], XGBoost [5], and LightGBM [11]; deep sequence models, including CNN-BiLSTM-Attention and Transformer [15]; and the period-aware backbone TimesNet [16]. All methods use the same preprocessing, window construction, and data splits. Classical baselines flatten each traffic window into a fixed-length vector, while deep models use the ordered window as input. Hyperparameters are selected on the validation set. For MPF-LiteIDS, the number of selected periods is set to $k = 4$.

4.3 Implementation Details

MPF-LiteIDS is implemented in PyTorch and trained with Adam [12]. Experiments are conducted on a workstation with an Intel Core i7-12700 CPU, 32GB RAM, and an NVIDIA GeForce RTX 3060 12GB GPU. Table 2 lists the main settings. Each experiment is repeated five times with different random seeds. Model complexity is measured by trainable parameters, FLOPs, model size, and average inference time.

Table 2. Main experimental settings.

Setting	Value
Optimizer	Adam
Learning rate	1×10^{-3}
Batch size	64
Epochs	50
Early stopping patience	5
Window length T	64
Window stride s	32
Selected periods k	4
Hidden dimension	128
Dropout rate	0.5
Self-distillation weight λ	0.3
Repeated runs	5

4.4 Evaluation Metrics

Accuracy, Precision, Recall, F1-score, and macro one-vs-rest AUC are used as the main evaluation metrics. For multi-class classification, macro-averaged Precision, Recall, and F1-score are reported to reduce the influence of class imbalance. The metrics are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (13)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (14)$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (15)$$

In intrusion detection, Recall and F1-score are particularly important because missed attacks may cause severe security risks. Therefore, F1-score is used as the primary metric in the following comparison and ablation analysis.

5 Results and Discussion

5.1 Overall Detection Performance

Table 3 presents the macro-averaged evaluation results. MPF-LiteIDS consistently delivers the highest F1-score and AUC across all three benchmarks, validating the efficacy of the multi-period traffic folding paradigm over generic time-series backbones like TimesNet [16]. Although minor dataset-specific metric fluctuations exist among baseline models due to diverse traffic distributions, MPF-LiteIDS achieves the best F1-score and AUC on all three datasets, indicating a more favorable balance between detection sensitivity and false-alarm control.

The performance gain is mainly attributed to two factors. First, multi-period folding exposes repeated traffic structures that are difficult to capture in the original one-dimensional sequence. Second, period-stage attention assigns adaptive weights to different periods and stages, allowing the detector to focus on attack-relevant traffic segments.

5.2 Ablation Study

To verify the contribution of each component, we conduct ablation experiments on BoT-IoT. Table 4 presents the detailed results. Compared with the full model, removing MPF leads to the largest F1-score decrease, from 97.81% to 96.76%, which indicates that period-aware representation is the key factor. Removing PSA decreases the F1-score to 97.22%, showing that adaptive period-stage fusion is useful for selecting discriminative traffic patterns. Removing SD reduces the F1-score to 97.52%, suggesting that self-distillation provides additional training regularization for the compact detector.

Table 3. Overall performance comparison on three datasets. F1-score is reported as mean $\pm$ standard deviation over five runs.

Dataset	Model	Acc. (%)	Pre. (%)	Rec. (%)	F1-score (%)	AUC (%)
BoT-IoT	RF	95.21	94.86	95.03	94.94 $\pm$ 0.15	97.12
	XGBoost	96.08	95.72	95.91	95.81 $\pm$ 0.11	97.84
	LightGBM	96.26	95.94	96.08	96.01 $\pm$ 0.09	97.96
	CNN-BiLSTM-Att.	96.71	96.43	96.58	96.50 $\pm$ 0.14	98.31
	Transformer	96.96	96.70	96.82	96.76 $\pm$ 0.12	98.47
	TimesNet	97.32	97.08	97.17	97.12 $\pm$ 0.10	98.71
	MPF-LiteIDS (Ours)	98.02	97.88	97.75	97.81 $\pm$ 0.08	99.02
ToN-IoT	RF	93.67	93.20	93.46	93.31 $\pm$ 0.18	96.12
	XGBoost	94.82	94.55	94.41	94.48 $\pm$ 0.13	96.91
	LightGBM	95.85	95.62	95.54	95.58 $\pm$ 0.11	97.74
	CNN-BiLSTM-Att.	95.78	95.44	95.60	95.52 $\pm$ 0.16	97.68
	Transformer	96.05	95.77	95.84	95.80 $\pm$ 0.12	97.91
	TimesNet	96.42	96.16	96.22	96.19 $\pm$ 0.09	98.18
	MPF-LiteIDS (Ours)	97.14	96.93	96.86	96.89 $\pm$ 0.07	98.54
CICIoT2023	RF	94.38	94.06	93.87	93.96 $\pm$ 0.17	96.58
	XGBoost	95.31	95.05	94.92	94.98 $\pm$ 0.12	97.22
	LightGBM	95.52	95.28	95.13	95.20 $\pm$ 0.10	97.36
	CNN-BiLSTM-Att.	96.25	96.04	95.93	95.98 $\pm$ 0.15	97.91
	Transformer	96.53	97.32	95.76	96.24 $\pm$ 0.13	98.31
	TimesNet	96.86	96.63	96.52	96.57 $\pm$ 0.11	98.33
	MPF-LiteIDS (Ours)	97.48	97.26	97.14	97.20 $\pm$ 0.09	98.66

Table 4. Ablation study on BoT-IoT.

Variant	Acc.	Rec.	F1	AUC
CNN-BiLSTM-Att.	96.71	96.58	96.50	98.31
w/o MPF	96.89	96.72	96.76	98.40
w/o PSA	97.31	97.16	97.22	98.63
w/o SD	97.64	97.45	97.52	98.78
Full model	98.02	97.75	97.81	99.02

Here, MPF denotes multi-period folding, PSA denotes period-stage attention, and SD denotes self-distillation. The results show a consistent improvement from the baseline to the full model, which supports the effectiveness of the proposed design.

5.3 Complexity Analysis

Table 5 compares the computational complexity of different deep models. FLOPs are calculated under the same input window length, and inference time is measured with batch size 1 on an NVIDIA GeForce RTX 3060 12GB GPU. Although Transformer and TimesNet achieve competitive performance, they introduce larger computational cost and higher inference latency. In contrast, MPF-LiteIDS preserves multi-period representation ability while using a lightweight convolutional extractor and compact period-stage attention. MPF-LiteIDS obtains the best F1-score while keeping the model size and inference time lower than CNN-BiLSTM-Attention, Transformer, and TimesNet. This result is important for IoT scenarios, where detection models are expected to be deployed under limited memory and computation.

Table 5. Complexity comparison on BoT-IoT.

Model	Params	FLOPs	Size	Time	F1
CNN-BiLSTM-Att.	2.14M	44.9M	8.6MB	2.13ms	96.50
Transformer	3.76M	71.5M	15.2MB	3.41ms	96.76
TimesNet	2.83M	58.4M	11.4MB	2.76ms	97.12
MPF-LiteIDS	1.36M	29.8M	5.5MB	1.47ms	97.81

Compared with CNN-BiLSTM-Attention, MPF-LiteIDS reduces parameters by 36.4% and FLOPs by 33.6%, while improving F1-score by 1.31 percentage points. This shows that the proposed lightweight design improves the accuracy-efficiency trade-off rather than merely increasing model capacity.

5.4 Attention Visualization

Fig. 3 visualizes the period-stage attention weights of different attack types. Each heatmap shows class-averaged normalized attention weights over correctly classified test samples. Rows denote selected dominant periods, and columns denote traffic stages. Darker cells indicate higher attention weights.

The visualization shows that different attacks activate different period-stage patterns. DDoS traffic tends to produce continuous high attention over the first two selected periods, reflecting persistent bursty traffic. DoS traffic presents stage-specific high responses, indicating that the attack intensity changes across traffic stages. Reconnaissance traffic shows more sparse and repeated responses, which is consistent with probing behavior. These results further support the motivation of modeling IoT traffic from a period-stage perspective.

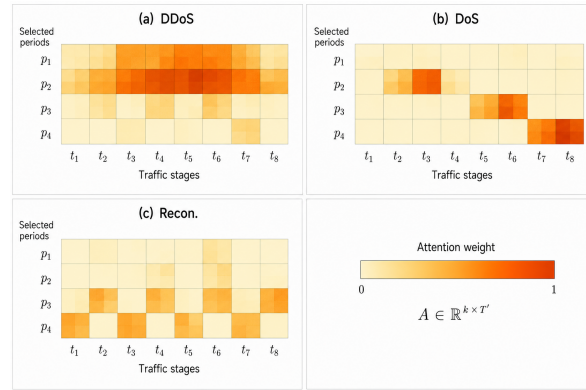


Fig. 3. Period-stage attention visualization.

6 Conclusion

This paper proposed MPF-LiteIDS, a multi-period lightweight attention network for IoT intrusion detection. Unlike conventional deep IDS models that mainly process traffic as a one-dimensional sequence, MPF-LiteIDS first identifies dominant frequency indices and folds traffic sequences into multiple two-dimensional periodic maps. This design enables the model to capture both intra-period attack bursts and inter-period behavioral shifts. Based on the folded representations, a lightweight detection network with depthwise separable convolution, BiLSTM encoding, and period-stage attention is designed to improve the accuracy-efficiency trade-off. In addition, self-distillation is introduced during training to enhance the compact model without increasing inference cost.

Experiments on three IoT intrusion detection datasets demonstrate that MPF-LiteIDS achieves higher detection performance than strong baselines, including Transformer and TimesNet, while maintaining lower model complexity. Ablation results further verify the effectiveness of multi-period folding, period-stage attention, and self-distillation. The attention visualization also shows that different attack types present distinct period-stage response patterns, supporting the motivation of period-aware traffic modeling. Future work will focus on on-line detection under streaming traffic, cross-device adaptation, and deployment-oriented optimization on real edge platforms.

References

1. Al-Garadi, M.A., Mohamed, A., Al-Ali, A., Du, X., Ali, I., Guizani, M.: A survey of machine and deep learning methods for internet of things (IoT) security. *IEEE Communications Surveys & Tutorials* **22**(3), 1646–1685 (2020). <https://doi.org/10.1109/COMST.2020.2988293>
2. Alsaedi, A., Moustafa, N., Tari, Z., Mahmood, A., Anwar, A.: TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems. *IEEE Access* **8**, 165130–165150 (2020). <https://doi.org/10.1109/ACCESS.2020.3022862>

3. Breiman, L.: Random forests. *Machine Learning* **45**(1), 5–32 (2001). <https://doi.org/10.1023/A:1010933404324>
4. Chaabouni, N., Mosbah, M., Zemmari, A., Sauvignac, C., Faruki, P.: Network intrusion detection for IoT security based on learning techniques. *IEEE Communications Surveys & Tutorials* **21**(3), 2671–2701 (2019). <https://doi.org/10.1109/COMST.2019.2896380>
5. Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 785–794 (2016). <https://doi.org/10.1145/2939672.2939785>
6. Chollet, F.: Xception: Deep learning with depthwise separable convolutions. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 1251–1258 (2017). <https://doi.org/10.1109/CVPR.2017.195>
7. Fawaz, H.I., Lucas, B., Forestier, G., Pelletier, C., Schmidt, D.F., Weber, J., Webb, G.I., Idoumghar, L., Muller, P.A., Petitjean, F.: InceptionTime: Finding AlexNet for time series classification. *Data Mining and Knowledge Discovery* **34**, 1936–1962 (2020). <https://doi.org/10.1007/s10618-020-00710-y>
8. Furlanello, T., Lipton, Z.C., Tschannen, M., Itti, L., Anandkumar, A.: Born-again neural networks. In: *Proceedings of the 35th International Conference on Machine Learning*. pp. 1607–1616 (2018)
9. Hinton, G., Vinyals, O., Dean, J.: Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531* (2015)
10. Howard, A.G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., Adam, H.: MobileNets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861* (2017)
11. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.Y.: LightGBM: A highly efficient gradient boosting decision tree. In: *Advances in Neural Information Processing Systems*. vol. 30, pp. 3146–3154 (2017)
12. Kingma, D.P., Ba, J.: Adam: A method for stochastic optimization. In: *International Conference on Learning Representations* (2015)
13. Koroniotis, N., Moustafa, N., Sitnikova, E., Turnbull, B.: Towards the development of realistic botnet dataset in the internet of things for network forensic analytics: Bot-IoT dataset. *Future Generation Computer Systems* **100**, 779–796 (2019). <https://doi.org/10.1016/j.future.2019.05.041>
14. Neto, E.C.P., Dadkhah, S., Ferreira, R., Zohourian, A., Lu, R., Ghorbani, A.A.: CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment. *Sensors* **23**(13), 5941 (2023). <https://doi.org/10.3390/s23135941>
15. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is all you need. In: *Advances in Neural Information Processing Systems*. vol. 30, pp. 5998–6008 (2017)
16. Wu, H., Hu, T., Liu, Y., Zhou, H., Wang, J., Long, M.: TimesNet: Temporal 2d-variation modeling for general time series analysis. In: *International Conference on Learning Representations* (2023)